



December 2025

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- 1-

11. Spring shackles are used to join...
 - A) Chassis frame and spring
 - B) Spring and Axle
 - C) Chassis frame and axle
 - D) All of the above
12. Which of the following is the disadvantage of the cone clutch?
 - A) It becomes difficult to disengage the clutch when the cone angle is less than 20°
 - B) It is silent in operation
 - C) The normal force on the contact surface is larger than the axial force
 - D) Same torque can be transmitted for the same size as the plate clutch
13. What is the primary function of the transmission in an automobile?
 - A) To direct exhaust gases away from the engine
 - B) To control the air-fuel mixture
 - C) To provide a means of connecting and disconnecting the engine to the driving wheels
 - D) To cool the engine
14. When an automobile takes a turn to avoid the skidding and over turning...
 - A) Each wheel will roll independently
 - B) Front two wheels will have a common centre during a turning
 - C) All the 4 wheels will roll on arcs having a common centre
 - D) None of these
15. Coil springs absorb shocks by...
 - A) Bending
 - B) Twisting
 - C) Compression
 - D) Tension
16. The metal used for the brake drum is...
 - A) Aluminium alloy
 - B) Cast steel
 - C) Pressed steel
 - D) Cast iron
17. In the case of brake linings, saturants and bonding materials are used to...
 - A) Facilitate moulding and curing of brake linings
 - B) To improve the heat dissipating capacity of linings
 - C) To give a constant coefficient of friction up to 800°C
 - D) To make the lining material impervious to moisture and oil and to ensure a steady coefficient of friction up to 3000°C
18. The most widely used brakes are operated...
 - A) Electrically
 - B) Hydraulically
 - C) By air pressure
 - D) By vacuum
19. Brake lining area required for shoe brakes for trucks is mainly based on...
 - A) Average truck speed
 - B) Frequency of brake applications
 - C) Gross vehicle weight
 - D) Material used for brake drums
20. Which of the following is one of the major exhaust emissions from CI engines compared to SI engines?
 - A) Oxides of nitrogen
 - B) Particulates
 - C) CO and CO_2
 - D) Unburnt hydrocarbon

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat sketch explain the constructional details of a piston.	8	L*2	1	1
	b) Explain the working of the battery ignition system.	8	L2	1	1
2.	a) What is the necessity of cooling system in an engine? Explain working of forced circulation system of water cooling with a neat diagram.	8	L2	1	1
	b) Explain wet sump lubrication system with a neat sketch.	8	L2	2	1

ME2001-1**SEE – December 2025**

- | | | | | | | |
|----|----|---|---|----|---|---|
| 3. | a) | Mention different types of combustion chamber designs used in spark ignition engine. | 8 | L2 | 2 | 1 |
| | b) | Mention the importance of ignition timing. Explain the working of vacuum advance ignition system. | 8 | L2 | 1 | 1 |

Unit – II

- | | | | | | | |
|----|----|---|---|----|---|---|
| 4. | a) | Explain the factors to be considered while designing the clutch. | 8 | L2 | 3 | 1 |
| | b) | List types of steering gear boxes used and explain any two with a neat diagram. | 8 | L2 | 4 | 1 |
| 5. | a) | Briefly explain the requirements of the clutch. | 8 | L2 | 3 | 1 |
| | b) | Explain the working of the air suspension system. | 8 | L2 | 4 | 1 |
| 6. | a) | Explain the working of torque converter with a neat diagram. | 8 | L2 | 3 | 1 |
| | b) | Explain the constructional details and working of the leaf spring with a neat sketch. | 8 | L2 | 4 | 1 |

Unit – III

- | | | | | | | |
|----|----|---|---|----|---|---|
| 7. | a) | Briefly explain the working hydraulic braking system. | 8 | L2 | 5 | 1 |
| | b) | Explain the importance of having EGR system in the engine and also explain the working of EGR system with a neat diagram. | 8 | L2 | 5 | 1 |
| 8. | a) | Write a short note on BS-VI and BS-IV. | 8 | L2 | 5 | 1 |
| | b) | Explain the working of crank case ventilation system. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
